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Limited Forearm Motion Compensated by Thoracohumeral Kinematics When Performing Tasks Requiring Pronation and Supination

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This study investigates the altered thoracohumeral kinematics when forearm rotation is restricted while performing five activities requiring pronation and supination. Two splints simulated both a fixed-supinated or fixed-neutral forearm in six healthy subjects; the three-dimensional coupled relationship among motion about the forearm, elbow, and shoulder were analyzed. In using a screwdriver, the normal range of forearm rotation of 77.6° (*SD* = 30.8°) was reduced in the fixed-supinated to 11.3° (*SD* = 2.9°) and fixed-neutral to 18.2° (*SD* = 6.2°). This restriction from the fixed-supinated and fixed-neutral forearms was compensated at the shoulder by a significant increase in the total range of (1) ad/abduction by 57.3° and 62.8° respectively ($p < .001$), (2) forward-reverse flexion (24.3° and 18.2° respectively; $p < .05$) and (3) internal-external rotation (37.1° and 44.2° respectively; $p < .001$). A similar result was demonstrated for the doorknob activity. The elbow did not significantly contribute to forearm rotation ($p = .14$), and is believed to be due to the elbow axis being orthogonal and oblique to the forearm axis. For open kinetic-chain activities, with a fixed-supinated forearm performing there was a significant coupled increase in ad/abduction ($p < .05$) and int/external rotation ($p < .05$) for the phone and feeding tasks, with the phone task also having a significantly increased forward shoulder flexion ($p < .05$). For the fixed-neutral forearm, significant compensatory movement was only seen in the feeding task with increased ad/abduction and internal-external shoulder rotation ($p < .05$) and the card inserting task with increased ad/abduction and forward-reverse shoulder flexion. Limited forearm function requires compensatory motion from adjacent joints to perform activities that require pronation and supination. This study quantifies the compensatory mechanism about the shoulder in a forearm limited in prosupination.

Keywords: activities of daily living, compensatory mechanisms, forearm biomechanics, limited forearm rotation, shoulder kinematics

Disorders to the forearm due to trauma, infection or pathology are disabling and affect the coordination of the entire upper limb, very often limiting the range of motion (Anglin & Wyss, 2000; Boone & Azen, 1979; Graham et al., 1998; Grood & Suntay, 1983; Kapandji, 2001). Surgical procedures in creating single bone forearm or forearm arthrodesis also sacrifices forearm rotation, while splints fixed in supination or neutral do immobilize a fractured forearm. The forearm that is restricted in motion is compensated by an increase in the kinematics about the adjacent joints of the upper limb (Carey et al., 2008; Morrey et al., 1981; Kapandji, 2001). This is because of the interjoint coordination system of the upper limb that spans from the thorax to the metacarpals which allow for several degrees of freedom that the upper limb can offer in a task (Crisco et al., 2007; Ishikawa et al., 2005; Van Andel et al., 2008). Several studies have highlighted compensatory kinematics from adjacent joints in restricted upper limb. In fractures involving the distal radius and distal radioulnar joint, patients commonly encounter posttraumatic restricted forearm motion and have been observed to use compensatory kinematics to perform various activities of daily living (Cooney et al., 1980; Linscheid, 1992). This has also been demonstrated in a study of hemiplegic cerebral palsy patients with impaired forearm rotation where compensatory trunk and shoulder movements are used (Kreulen et al., 2007). In a review of cases of radioulnar synostosis, reference is made to the fatiguing of the extremities when the shoulder is used

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to perform activities requiring pronation and supination (Kasten et al., 2009). They have cautioned that compensating for forearm rotation could make the shoulder prone to overuse, and this could be more disabling. Yet there are limited data suggesting or quantifying the compensatory relationship between the range of motion about the shoulder and the forearm rotation, particularly in performing various activities of daily living requiring pronation and supination of the hand. Most studies of the forearm look at just the normal range of forearm rotation and their range of limitation without associating it to any compensatory motion from the proximal adjacent joints (Anglin & Wyss, 2000; Boone & Azen, 1979; Nakamura et al., 1999; Safaee-Rad et al., 1990; Tay et al., 2008).

The study therefore investigated the range of forearm motion to perform various unilateral tasks that required obligatory forearm rotation and determined the degree of compensation by the coupled kinematics about shoulder and elbow, when the forearm and wrist rotations were restricted. Our interest in doing this study was to simulate limited forearm rotation and wrist movement associated with posttraumatic forearm injuries, as we wanted to establish a measure to monitor the performance of posttraumatic forearm injuries during recovery. We were concerned about the possible risk of soft tissue injuries at the adjacent joints where expected increases in shoulder and elbow range of motion and usage could compound the disability of the upper limb (Carey et al., 2008; Cooney et al., 1980; Graham et al., 1998; Linscheid, 1992; Mell et al., 2005; Van Andel et al., 2008). Distally, the motion of the forearm is intimately linked to that of the wrist, and makes it a challenge to separate the kinematics about their joint axes (Adams et al., 2003; Crisco et al., 2007; Ishikawa et al., 2005; Van Andel et al., 2008). We therefore chose to limit the wrist motion to allow us to specifically study how limited forearm rotation is compensated by the proximally adjacent joints. To reduce other variabilities, we also limited to possible compensatory contributions from the torso.

From clinical observations the obvious hypothesis that we tested was that when the forearm is restricted in its range of forearm rotation, the loss of movement to perform unilateral activities requiring pronation and supination is compensated by increased thoracohumeral and elbow articulations. What is not known is the degree of compensation and usage, and a quantitative measure of the altered kinematics.

Methods

Three males and 3 females (mean age 23.8 years old, *SD* 0.98) with no history of musculoskeletal disorders were enrolled in the study (IRB-05–128). All were right-hand dominant, and were instructed to perform various tasks using their right hand. The experiment was explained in detail to the subjects and their questions satisfactorily answered.

Subjects were first asked to perform five unilateral ADLs, without any restriction to forearm rotation. These were as follows. ADL-I: Turning a doorknob. The subject was to complete a full cycle (range: 70 degrees) by turning the doorknob clockwise (+) and then counter-clockwise (–), beginning from a fully pronated position. ADL-II: Using a screwdriver. Subjects completed a full cycle (range: 120 degrees) turning the screwdriver to tighten a screw on the board, clockwise (+), and then counter-clockwise (–), beginning from a fully pronated position. ADL-III: Answering the telephone. Subjects completed a full cycle by picking up the phone handset and putting the earpiece to their ear. ADL-IV: Feeding oneself with a spoon. Subjects completed a full cycle by using the spoon to scoop up some cereal from a bowl and bringing the spoon from the bowl to their mouth (+), then returning the spoon to the bowl (–). ADL-V: Taking a card from the shirt pocket and inserting it into a card slot. Subjects completed a full cycle by reaching into the shirt pocket to take out a card, and then inserting it into a card reader placed in front of the subject.

The doorknob and screwdriver tasks were considered closed kinetic chain activities, as the hand held on to a fixed object; the phone, spoon feeding and card slotting activities were considered open kinetic chain activities. The subjects were made to sit on a chair with a backrest, and with both feet on the ground. The seat height was adjusted so that each subject had their elbows comfortably flexed at 90 degrees on a table with shoulder at 0 degrees of flexion. A setup comprising a doorknob, a screw and a card slot were mounted onto a wooden board and placed about 50 cm in front of the midline of the subject. A telephone and a cereal bowl with a spoon were also placed at the same position during the experiment. The subjects were instructed to start with their hand on the table (with forearm in neutral position) but they were free to choose their own path in performing each of the ADL. For each task the subjects were asked to do three trials. For all tasks, the subjects were allowed to practice performing the task before the actual trial. The subjects were advised to ensure that body sway was minimized. All subjects were given the same instructions and were not told how to compensate for the loss of forearm function.

Limited forearm rotation was simulated by wearing a forearm brace. Two standard thermoplastic splints were made by a trained Hand Therapist. One restricted the forearm in a fixed-supinated position while the other limited the rotation of the forearm in a fixed-neutral position (Figure 1). The splints allowed for elbow flexion but limited radial/ulnar deviation and flexion-extension at the wrist. Three experimental groups were compared. The first was the normal forearm where the five unilateral ADLs were done with the forearm unrestricted. The subjects then repeated the tasks wearing the fixed-supinated, first, and subsequently wearing the fixed-neutral splints, for all cases. The same two splints were used for all subjects.

In our study we chose to model the kinematics of the forearm with the radius and ulna as two separate segments, allowing us to measure the articulation of radius with respect to the ulna. This will become relevant when we want to study disorders in the forearm where we will need to delineate the kinematics between the radius and ulna, and between the ulna and humerus.

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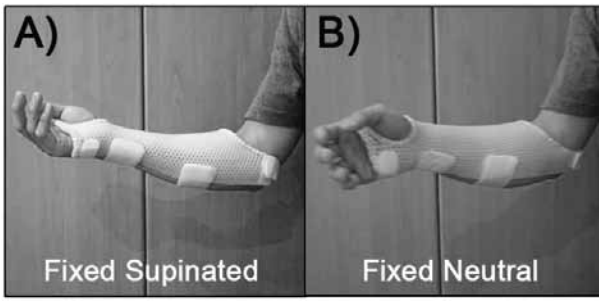


Figure 1 — Thermoplastic splints used to simulate a (A) fixed supinated forearm and a (B) fixed neutral forearm in a normal health subject.

The kinematics of the upper limb for each activity were observed using a 5-camera Vicon motion analysis system (Model VICON 460, Vicon Peak, Oxford, United Kingdom). Comparisons were made between the normal condition and while donning either of the splints. The orientation of the upper limb was determined based on the thoracohumeral movement, with the global coordinate system on the thorax (upper torso). The kinematics of the scapula (scapular-thoracic movement) was a challenge to measure due to excessive skin movement and was excluded, and became a limitation in this study. Spherical retro-reflective 15 mm markers placed on anatomically well-defined body landmarks defined the various segments of the upper limb and torso (Table 1, Figure 2). These same landmarks were used to define the coordinate

Table 1 Marker placements on bony landmarks and the local coordinate systems of the various segments

Markers Used on Bone/Body Landmarks	Local Coordinate Systems
Body (Thorax) Segment	$\mathbf{O}_T$: the origin coincident with IJ
Right acromion (AcrR) Left acromion (AcrL)	$\mathbf{Y}_T$: The line connecting the midpoint between PX and T10, and the midpoint of IJ and C7 in the superior direction
Process spinosus cervical 7 (C7) Incisura jugularis or jugularis notch (IJ)	$\mathbf{Z}_T$: The line perpendicular to the plane formed by IJ, C7, and the midpoint between PX and T8, pointing to the right
Process xiphoideus (PX) Process spinosus thoracic 8 (T8)	$\mathbf{X}_T$: The common line perpendicular to the Z_T- and Y_T-axis, pointing forward
Humerus Segment	$\mathbf{O}_H$: origin coincident with GH
Medial epicondyle (EM)	$\mathbf{Y}_H$: The line connecting GH and the midpoint of EL and EM, pointing to GH
Lateral epicondyle (EL)	$\mathbf{X}_H$: The line perpendicular to the plane formed by EL, EM, and GH, pointing forward
Glenohumeral rotation center (GH), which is taken approximately 7 cm inferior to the AcrR	$\mathbf{Z}_H$: The common line perpendicular to the Y_H- and X_H-axis, pointing to the right
Ulnar Segment	$\mathbf{O}_U$: The origin coincident with US
Ulnar styloid (US)	$\mathbf{Y}_U$: The line connecting US and OLE0 pointing to OLE0
Olecranon process (OLE0) Wand marker attached to OLE0 (OLE1)	$\mathbf{Z}_U$: The line parallel to line connecting OLE0 and OLE1, when the radius is in neutral forearm rotation, pointing laterally away from the RS
	$\mathbf{X}_U$: The common line perpendicular to the Y_U- and Z_U-axis
Radial Segment	$\mathbf{O}_R$: The origin coincident with RS
Radial styloid (RS)	$\mathbf{Y}_R$: The line connecting RS and RS1 pointing to RS1
Wand marker attached to the RS (RS0) 5 cm proximal to RS along the radius (RS1)	$\mathbf{X}_R$: The line perpendicular to the plane formed by RS, RS0, and RS1, pointing toward RS0
	$\mathbf{Z}_R$: The common line perpendicular to the Y_R- and X_R-axis, pointing laterally
Dorsal Hand Segment	$\mathbf{O}_D$: The origin coincident with midpoint of M2 and M5
Dorso-distal end of the 2nd metacarpal (M2)	$\mathbf{Y}_D$: The line connecting the midpoint of M2 and M5 and M3, and pointing to M3
Dorso-distal end of the 5th metacarpal (M5) Dorso-proximal end of the 3rd metacarpal (M3)	$\mathbf{Z}_D$: The line perpendicular to the plane formed by M2, M5, and M3, pointing laterally
	$\mathbf{X}_D$: The common line perpendicular to the Y_D- and Z_D-axis, pointing to the thumb

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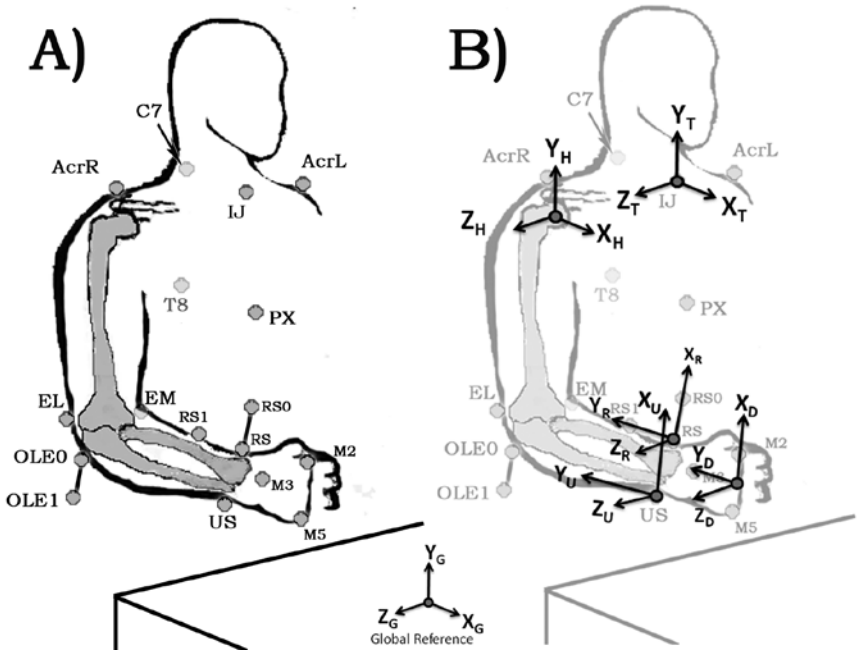


Figure 2 — Schematics of the (A) anatomical landmarks and marker placements and (B) the local coordinate systems of the thorax, humerus, ulna, radius and hand (see Table 1 for legend).

system when the splints were donned on as well. Some of the optical markers had to be removed when the splints were put on. These markers were later reattached. The selection of bony landmarks took into consideration the need to minimize skin movement (Anglin & Wyss, 2000; Wu et al., 2005). For the elbow, the joint motion was described as the rotation of the ulna with respect to the humerus. For forearm motion, the rotation of the radius with respect to the ulna was used. For wrist motion, we described the motion of the hand, defined by the markers placed on the dorsal side of the hand with respect to the radius. We applied the floating axis equations to determine a joint coordinate system (JCS) for the thoracohumeral, elbow, forearm and wrist rotations (Figure 3). The JCS for the thoracohumeral and the forearm rotation differed from the ISB recommendations (Wu et al., 2005) to suit the clinical descriptions of the joint rotations used in our laboratory. The joint coordinate system is described by three unit vectors e_1, e_2 and e_3. The body fixed axes, e_1 and e_3, describe the respective fixed and moving segments across each joint, while the reference axes, e_1^r and e_3^r, describe the axes perpendicular to the body fixed axis for each respective segment. The reference axis, e_1^r, coincided with the long axis of the proximal segment while the fixed axis, e_3 coincided with the long axis of the distal segment. The exceptions were in the case of the thoracohumeral rotation, where the body fixed axis, e_1 coincided with the Y_T-axis of the thorax coordinate system; and in the case of forearm rotation where the reference axis, e_3^r, in the distal segment coincided with the X_R-axis of the radius coordinate system. The common (floating) axis, e_2 was perpendicular to both the body fixed axes, with its orientation and relative position given by

$$e_2 = \frac{e_3 \times e_1}{|e_3 \times e_1|}$$

The magnitudes of the rotations at each joint were then measured by the angles α and γ, about the body fixed axes, e_3 and e_1 respectively, while angle, β, was the rotation about the floating axis, e_2.

We chose to deviate from the ISB recommendations for the JCS of the various upper limb joints, to allow the description of each of the orthogonal rotations, α, β and γ, to coincide with the clinical rotation descriptions (Figure 3).

The motion analysis data were postprocessed using *Bodybuilder for Biomechanics*, labeling the moving markers and estimating the relative angles between segments at 50 Hz.

For each activity, the three groups (normal, fixed-supinated and fixed-neutral) were compared for their mean total range of motion using a repeated measures analysis of variance (ANOVA) at a 0.05 level of significance. A Scheffé post hoc test determined the difference between the activities and groups. The types of activities were categorized as either closed-chain activities (doorknob, screwdriver) or open-chain activities (phone, feeding and card inserting). Angle–angle relationship curves were plotted to demonstrate the changes to the shoulder rotations, between the fixed-supinated and the fixed-neutral forearms. These plots demonstrated the relationship between the shoulder motions and the forearm rotation in performing a task. When comparing the relationship between the groups, a steeper slope indicates that for every unit degree of forearm rotation, more shoulder rotation was required.

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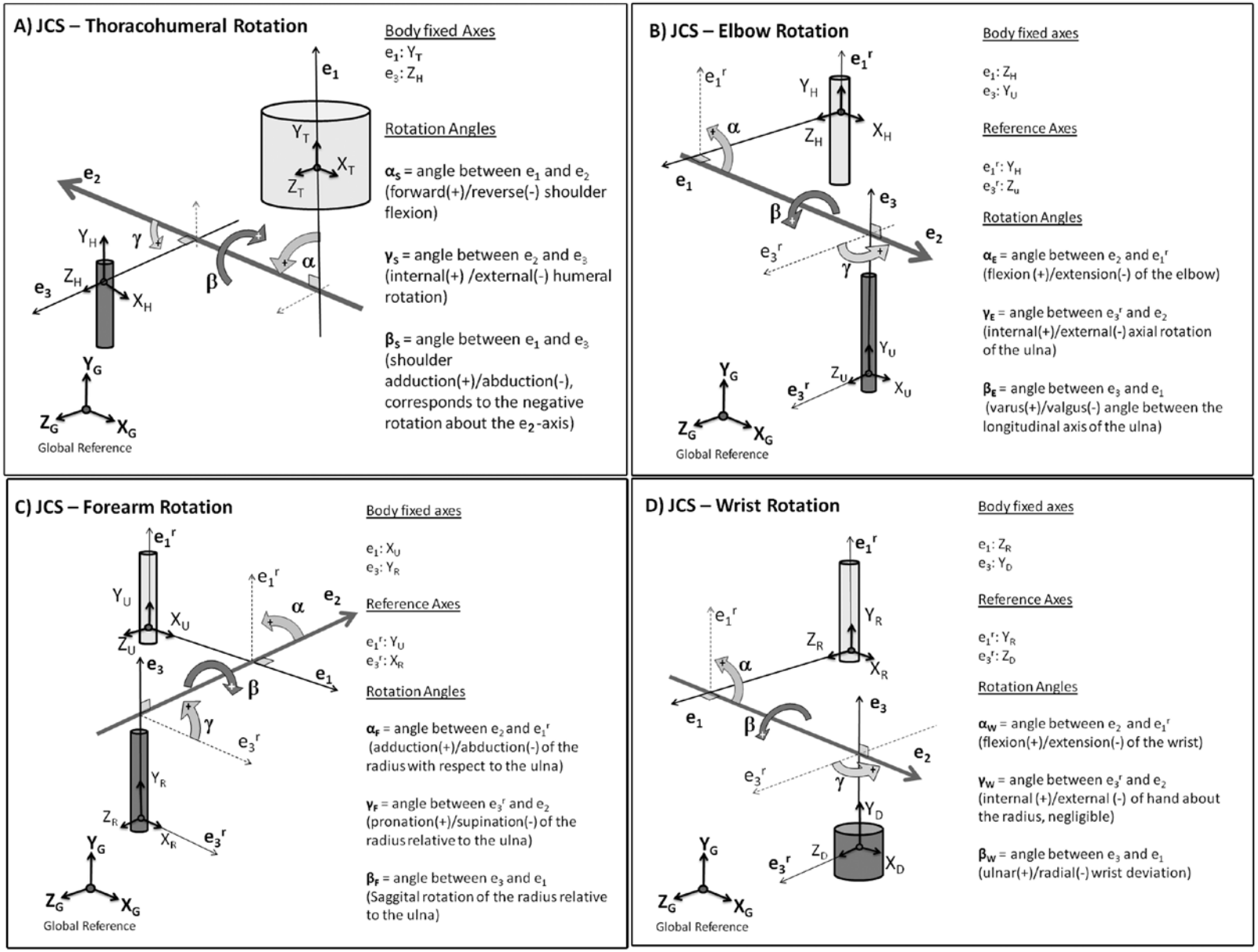


Figure 3 — The joint coordinate system for (A) Thoracohumeral rotation, (B) Elbow rotation, (C) Forearm rotation, and (D) Wrist rotation. The right-hand rule rotation about each of the axes e_1, e_2, and e_3 are defined by the angles gamma (γ), beta (β) and alpha (α) respectively.

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Results

The resting forearm position was measured with the splints in place, where the elbow was in a 90 degrees flexed position, with 0 degrees abduction and 0 degrees flexion at the shoulder. For the supinated-fixed splint the resting position was supinated at 36.7 (*SD*, 1.5) degrees, while donning the fixed-neutral splint the resting position was slightly supinated at 7.2 (*SD*, 0.8) degrees.

The mean total range of joint motions about the wrist, forearm and shoulder, in performing each of the five tasks in the two restricted forearm groups were compared with that of the normal forearm (Table 2).

In our study the mean range rotation was a measure of the articulation of the radius with respect the ulna. To qualify if our data were comparable to other published studies, the unrestricted mean range of forearm motion required to perform the doorknob, telephone and feeding tasks were compared against similar previously published data (Morrey et al., 1981) and found to differ in only the feeding activity (Figure 4).

The screwdriver task was used as a reference to assess the loss in range of forearm motion and the thoracohumeral compensation after the splints were put on. The required range of forearm rotation to perform to this task without the use of the splints, was 77.6 (*SD*, 30.2) degrees. When the splints were subsequently donned, forearm rotation for the same task was significantly limited to a mean range of 11.3 (*SD*, 5.9) degrees ($p < .001$) for the fixed-supinated forearm and a mean range of 18.2 (*SD*, 6.2) degrees ($p < .001$) for the fixed-neutral forearm.

In the open chain activities, the splints reduced the range of wrist flexion-extension, significantly (telephone task, $p = .001$; feeding task, $p = .001$; card-slotting task, $p = .03$). For the closed chain activities, the normal range of wrist flexion-extension was small (doorknob task, 23.9°; and screwdriver task, 25.7°), with no significant changes noted in the wrist flexion-extension range when the splints were put on (doorknob task, $p = .31$; screwdriver task, $p = .12$). The splinted forearms also resulted in less wrist radial-ulnar deviation, but this was only significant in the telephone task ($p = .001$) and the feeding task ($p < .001$). With or without donning the splints, there were no significant changes to the total range of elbow motion ($p = .14$) in performing all five activities (doorknob, $p = .29$; screwdriver task, $p = .126$; telephone task, $p = .69$; feeding task, $p = .32$; and card slotting task, $p = .55$).

Significant changes ($p < .05$) were recorded in the range of motion about the shoulder to perform each of the five tasks, between the fixed-forearm positions and the normal forearms (Figure 5).

For the closed-chain activities, a greater range of shoulder adduction-abduction was noted for both restricted forearms (doorknob task, $p = .01$; screwdriver task, $p < .001$). For the open-chain activities, the fixed-supinated forearms showed significantly higher total range of adduction-abduction in the telephone task ($p < .05$) and the feeding task ($p = .03$); while for the fixed-neutral forearms significantly higher total range of adduction-abduction were noted in the feeding task ($p = .03$); and card slotting task ($p = .03$).

For the closed chain activities, a greater total range of shoulder forward-reverse flexion was noted only in the screwdriver task, ($p < .05$) with no significant change in the door-knob task ($p = .62$). In the open chain activities, fixed-supinated forearms had a significantly greater range of forward-reverse flexion in the telephone task ($p = .04$) only; while the fixed-neutral forearms showed significant changes in the range of shoulder flexion in the card-slotting task ($p = .04$) only.

For the closed chain activities, a greater range of shoulder internal-external rotation was noted for both restricted forearms in the doorknob task ($p < .001$) and the screwdriver task ($p < .001$). For the open chain activities, only the fixed-supinated forearms showed significantly higher range of internal-external rotation in the telephone task ($p < .05$) and the feeding task ($p < .05$). The fixed-neutral forearms did not demonstrate any significant changes in internal-external rotation for any of the tasks.

The screwdriver task when performed normally required the largest range of pronation-supination, with the lowest range of wrist and elbow rotation. This was used to demonstrate the change in the motion about the shoulder in the two fixed forearm positions (Figure 6). This would not necessarily be representative of the other four tasks, as each would have involved the other adjacent joints with their respective articulations when performing the task.

To turn the screwdriver clockwise with the right hand (i.e., to supinate the hand) in the screwdriver task (0–50% cycle), the subjects had to compensate this by first abducting and internally rotating to initiate the task for both fixed forearm positions. To turn the screwdriver counterclockwise (i.e., to pronate) with the right hand (50–100% cycle) the arm began in an abducted, reversed flexion and externally rotated position about the shoulder, before initiating the task.

To supinate and pronate with the fixed forearm splints, the subjects had to use more shoulder articulation if the forearm was fixed in a more supinated position, as shown in the angle-angle relationships (Figure 7). For the screwdriver task the angle-angle relationships had good linear correlation with coefficient of determination, R^2 greater than 0.7, making the slope a good predictor for increased involvement of the shoulder to compensate with loss in forearm rotation. The fixed forearm positions required an increased adduction (the slopes for the fixed supinated, fixed-neutral, and normal were 10.66, 6.03 and –0.32, respectively), increased reverse flexion (–5.20, –2.75, 0.22, respectively) and increased external rotation (–5.38, –3.39, 0.13, respectively) about the shoulder to supinate the hand.

Discussion

The total normal forearm rotation range required to perform most activities of unilateral tasks is said to be

Table 2 Mean total range of motion at the (A) wrist, (B) forearm, (C) elbow and (D) shoulder for the five tasks, based on three trials for each task, performed with a normal forearm, fixed supinated forearm and fixed neutral forearm (*N* = 6 subjects)

	Activities				
	I	II	III	IV	V
	Doorknob	Screwdriver	Phone	Feeding	Card
Groups	**Mean (*SD*) range of motion (degrees)**				
A1. Wrist—flexion/extension (Mauchly's sphericity test not met, $p < 0.001$)					
Normal	25.7 (20.3)	23.9 (8.9)	47.1 (20.4)	43.6 (18.4)	56.4 (27.6)
Fixed Supinated	14.6 (10.7)	23.9 (20.6)	13.7 (5.1)#	18.8 (9.1)#	10.5 (4.0)#
Fixed Neutral	14.6 (6.8)	10.4 (7.5)	22.8 (4.4)#	13.2 (3.9)#	23.4 (18.5)#
A2. Wrist—radial/ulnar deviation (Mauchly's sphericity test met, $p = 0.36$)					
Normal	20.1(11.4)	15.4 (3.7)	64.3 (15.0)	35.1 (15.7)	41.9 (16.5)
Fixed Supinated	15.1 (7.7)	20.5 (11.2)	19.1 (13.7)#	20.9 (21.7)	10.1 (4.4)#
Fixed Neutral	16.8 (6.0)	10.0 (3.7)	28.2(21.1)#	15.7 (6.9)	13.8 (4.2)#
B1. Forearm—pronation/supination (Mauchly's sphericity test not met, $p = 0.001$)					
Normal	54.2 (24.5)	77.6 (30.8)	41.9 (5.6)	48.1 (11.9)	69.4 (18.8)
Fixed Supinated	10.9 (5.1)#	11.3 (5.9)#	13.1 (3.6)#	8.5 (4.3)#	18.4 (8.2)#
Fixed Neutral	15.8 (5.5)#	18.2 (6.2)#	15.2 (5.8)#	14.2 (5.2)#	15.6 (7.6)#
C. Elbow—flexion/extension (Mauchly's sphericity test not met, $p = 0.001$)					
Normal	9.0 (3.6)	7.0 (2.5)	21.8 (11.9)	29.5 (21.9)	27.1 (8.6)
Fixed Supinated	26.6 (28.3)	9.3 (5.8)	27.6 (17.6)	16.5 (9.8)	32.2 (13.8)
Fixed Neutral	22.5 (14.7)	14.7 (8.6)	20.5 (12.6)	21.2 (7.2)	25.8 (5.3)
D1. Shoulder—abduction/adduction (Mauchly's sphericity test met, $p = 0.20$)					
Normal	48.5 (30.3)	31.3 (15.9)	11.0 (7.0)	11.8 (3.4)	27.8 (13.3)
Fixed Supinated	101.0 (42.7)#	88.6 (30.0)#	40.3 (24.9)#	29.8 (16.0)#	53.8 (23.1)
Fixed Neutral	104.5 (7.0)#	94.1 (8.7)#	34.6 (23.7)	29.3 (12.0)#	71.5 (33.4)#
D2. Shoulder—forward/reverse flexion (Mauchly's sphericity test met, $p = 0.52$)					
Normal	34.8 (25.1)	21.8 (11.0)	14.4 (4.4)	12.1 (6.2)	34.8 (21.1)
Fixed Supinated	51.3 (39.2)	46.5 (28.4)	21.4 (4.3)#	17.2 (5.5)	61.8 (24.6)
Fixed Neutral	46.6 (22.2)	49.0 (13.0)#	16.8 (4.1)	20.3 (5.7)	79.9 (34.1)#
D3. Shoulder—internal/external rotation (Mauchly's sphericity test met, $p = 0.69$)					
Normal	15.4 (7.1)	12.8 (6.8)	23.8 (10.3)	19.5 (7.2)	33.8 (8.2)
Fixed Supinated	53.8 (17.1)#	49.9 (9.0)#	44.3 (12.8)#	32.7 (6.8)#	51.0 (25.4)
Fixed Neutral	64.3 (13.9)#	57.0 (13.1)#	33.2 (16.7)	27.7 (11.1)	43.4 (14.2)

Note. Repeated-measures ANOVA was used to determine the difference between the normal, fixed-supinated, and fixed-neutral forearms performing the five tasks, for each of the joint articulations. The Mauchly's sphericity test was used to determine homogeneity of the variance. Statistics: #$p < 0.05$, compared with the normal (one-way ANOVA) for each task.

between 75 degrees pronation and 85 degrees supination (Boone & Azen, 1979; Morrey et al., 1981). When there is a loss in the range of forearm motion, there are compensatory movement strategies involving other joints that will allow the task to still be carried out. This can come from motion about the wrist distally or proximally about the elbow, shoulder, as well as the trunk (Boone & Azen, 1979; Buckley et al., 1996; Carey et al., 2008; Kasten et al., 2009; Kreulen et al., 2007; Mell et al., 2005).

This study investigated the compensatory mechanisms from the coupled motion about the shoulder and elbow as a result of the forearm and wrist ranges of motion being restricted or limited. The splints limited the wrist position allowing us to better understand the contributions of the proximal adjacent joints. As we only wanted to assess the shoulder and elbow compensatory motion, we specifically instructed the subjects to avoid any trunk sway when performing the 5 tasks. This was done by instructing

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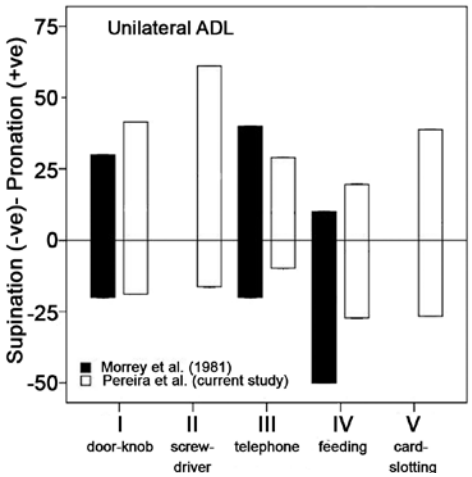


Figure 4 — The mean range of forearm rotation for the intact, unsplinted forearm required to perform 5 tasks were compared against data previously published by Morrey et al. (1981).

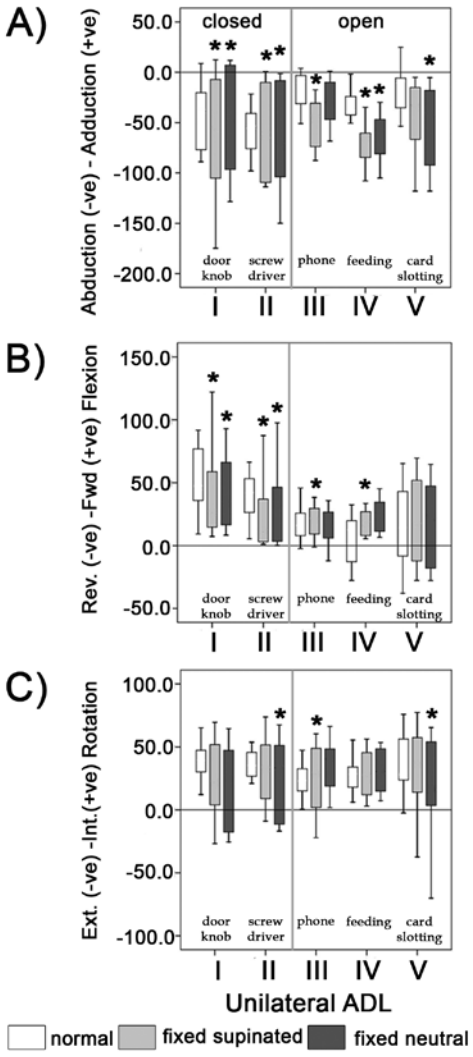


Figure 5 — Box-plots of the range of motion in (A) abduction-adduction of the shoulder; (B) reverse-forward flexion of the shoulder and (C) external-internal rotation of the shoulder (N = 6 subjects) while performing 5 tasks. The data compares without any splints (white bar), and then subsequently wearing a splint to simulate a fixed supinated forearm (light gray bars) and a fixed neutral forearm (dark gray bars). The five ADLs were categorized as closed-kinetic chain activities (I-doorknob, II-screwdriver) and open chain activities (III-telephone, IV-feeding, V-card inserting). For each activity, statistical significance (ANOVA, post hoc Scheffé) at $p < .05$ against the normal range for each task, is indicated with an asterisk (*).

them to keep their left arm fixed on the table while their backs were pressed against the chair rest. Previous work on understanding compensatory movements of transradial prosthesis users found that when trunk movement was not limited, compensatory kinematics from the shoulder was not found to be significantly different from normal subjects (Carey et al., 2008). By limiting torso movement, the assumption was that we could unmask the contributions from trunk movements, and specifically focus on understanding the compensatory shoulder and elbow kinematics in forearm limited in their range of motion.

The normal range of forearm rotation measured in this study for the 5 ADL tasks when compared with the data from Morrey et al. (1981) was no different for the doorknob and telephone tasks. There was however, a difference in the ranges obtained for the feeding task but the explanation was the difference in feeding techniques, as Morrey's data were based on feeding with a fork and that required more supination, compared with feeding with a spoon in our task (Morrey et al., 1981; Safaee-Rad et al., 1990). Also, Morrey's study measured the joint angles and the total range of motion using a goniometer.

In simulating a fixed range of forearm rotation, the two splints used allowed free elbow flexion and extension, but limited wrist motion to some extent with some degree of wrist movement observed. Previous work by others found that with partially or fully limited wrist motion, no functional differences were recorded in their range of motion to perform unilateral hand and wrist functions (Adams et al., 2003). This may support the argument that restricting wrist motion may not have influenced the compensatory thoracohumeral and elbow kinematics observed. We also limited movement in the torso. However, the work by Carey et al. (2008) found that when they included torso movement in their study to determine the compensatory movement of transradial prosthesis users during common tasks, no significant differences in shoulder compensations was noted when performing unilateral hand functions (Carey et al., 2008). Although this may support our argument to restrict torso movement, where our data might be useful for the development of a kinematics upper limb model, further studies are recommended to reassess this. Clinically, given that we may have created an unnatural condition of limiting wrist and torso movement, we do caution that it remains uncertain if the thoracohumeral compensation measured

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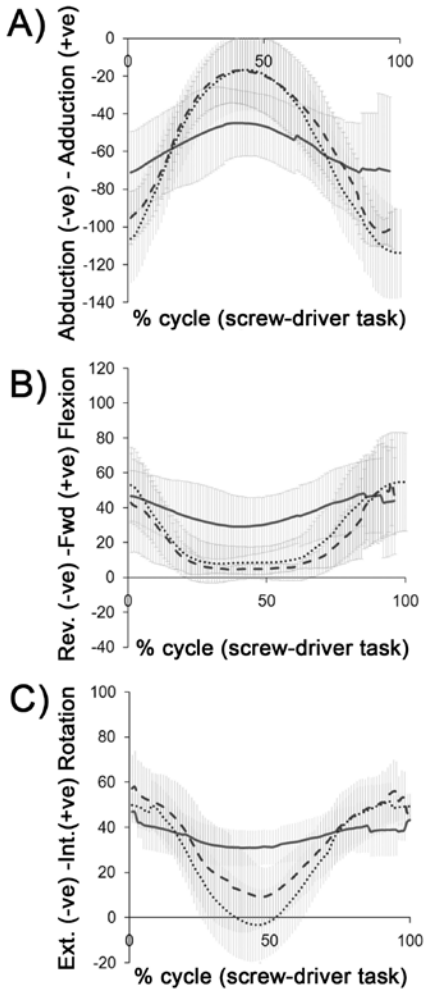


Figure 6 — Profiles of abduction-adduction; reverse-forward flexion and external-internal rotation of the shoulder against a full cycle of using a screwdriver with the right hand. The activity was first done without any splints (bold line), then subsequently with a splint simulating a fixed supinated (dotted line), and a fixed neutral forearm (dashed line). Each profile represents the mean of 6 subjects over a full cycle of the activity. The whiskers represent the standard deviation of each percentage point on the full cycle for each of the profiles. *Note.* 0–50% cycle: turning the screwdriver clockwise; 50–100%: turning the screwdriver counterclockwise).

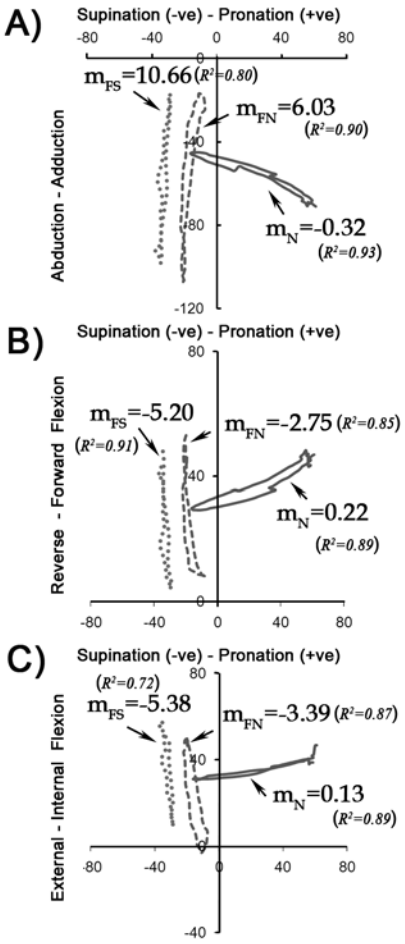


Figure 7 — A representative angle-angle relationship between forearm rotation (supination-pronation) and the three orthogonal rotation axes about the shoulder—(A) Abduction-adduction, (B) reverse-forward flexion and (C) external-internal rotation of the shoulder; required to perform the screwdriver task. *Note.* *m* = slope of the regression line representing the angle-angle relationship plots, and R^2= the coefficient of determination; N = normal (bold line), FS = fixed supinated (dotted line), and FN = fixed neutral (dashed line).

was due to the limited forearm rotation or the combination of limiting the forearm and wrist as well as the torso.

We used a common splint for all subjects, which allowed for some degree of wrist motion when the splints were donned. Unfortunately, we did not measure this "allowable" range of wrist motion in the cohort of subjects, which may have varied due to different hand sizes. These variations in the range of wrist motion made it uncertain if it influenced the coupled elbow and thoracohumeral kinematics that was measured. Nonetheless, from reviewing the different unilateral tasks performed irrespective of whether wrist motion was significantly restricted or not, significant altered thoracohumeral kinematics were recorded.

Between the closed and open kinetic chain activities, more wrist motion was observed in the open chain activities, and this was significantly restricted in simulated restricted forearms. For the closed chain activities, wrist motion was less in the unrestricted forearm, with no significant changes in wrist motion in the restricted forearm. The closed chain activities were therefore more independent of wrist compensation and gave a better representation of the shoulder and elbow compensation in the restricted forearms. In addition, since both the open- and closed-chain activities did not show any significant involvement of the elbow, much of the compensation resulted from motion involving the shoulder

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joint. We acknowledge that the study was limited by the small sample size. The Mauchly's sphericity test for wrist flexion-extension, pronation-supination, and elbow flexion-extension was not met and therefore heterogeneity of variance was considered. However, the Mauchly's test was met for the radial-ulnar deviation, and the shoulder articulations and we therefore assumed homogeneity of variances for these data sets.

At the elbow, the low of range motion recorded was perhaps limited due to the type of tasks, which were all performed while sitting in front of a table and within reach (40–50 cm). Elbow motion generally did not differ before and after the splints were put on. One other possible reason is that the axis of the elbow has an orthogonal orientation relative to the forearm rotation axis. The elbow would also generally be orthogonal to the altered axis of rotation if the splints were donned. So in either case, the involvement of the elbow in unilateral activities that required pronation and supination, would be minimal and their differences between the groups not significant as noted in the data.

For fixed-supinated forearms (Table 2, Figure 7), the total range of shoulder abduction-adduction and internal-external rotation was significantly increased for all ADL tasks, except the card slotting task which recorded a large variation in the total range of motion among the subjects. The total range of forward-reverse flexion was only significantly increased for the telephone task, whereas for all other tasks, they were not significantly different. For fixed-neutral, significantly higher total range of motion was noted in abduction-adduction and internal-external rotation, but this was only in the closed-chain activities (doorknob and screwdriver tasks). For the open-chain activities, the increases in the total range of motion in abduction-adduction were only seen in the feeding task and card slotting task Between fixed-supination and fixed-neutral, no clear significant difference was noted between the various task, suggesting that the altered mechanics was more related to the loss in forearm function rather than the type of activity.

Although the splints were intended to limit the forearm in a fixed-forearm position (Table 2), some degree of forearm rotation (i.e., the radius rotating about the ulna) was measured. The relations between the change in shoulder articulation with respect to a full cycle of each task as shown in Figure 6, provides a quick comparison of the amount of shoulder rotation in the restricted forearms normalized to the unrestricted forearm rotation. To establish an index for comparison between the three forearms, the angle-angle relationship was plotted between the shoulder movements against the forearm movement with the screwdriver task as an illustration (Figure 7). The slope of the regression line, m, of the angle-angle relationship gave us a measure of the increase in motion about the shoulder joints per unit degree of forearm rotation, allowing us to compare a set of normal forearms to set of disordered forearms. In using the screwdriver, we noted that abduction-adduction which was minimally involved in the normal forearm (m = –0.32) increased several folds with the restricted forearm positions (m = 6.03 for fixed-neutral and m = 10.66 for fixed-supination). The same were seen, but to a lesser extent for forward-reverse flexion (m = –2.75 for fixed-neutral and m = –5.20 for fixed-supination) and internal-external rotation (m = –3.39 for fixed-neutral and m = –5.38 for fixed-supination) of the shoulder. This demonstrated a trend that a forearm fixed in a more supinated position required more abduction-adduction, more forward-reverse flexion, and more internal-external rotation to compensate for the loss of forearm rotation.

In general, a combination of an increased shoulder adduction, reverse flexion and external humeral rotation would compensate for a loss in forearm supination, while a combination of increased shoulder abduction, forward flexion and internal humeral rotation would compensate for a loss in forearm pronation.

There was some degree of hysteresis recorded in the angle-angle relationships, without and with the splints put on. This study did not determine the cause of this, but we postulate that the likely source is either due to the translational shift of the humeral head with respect to the torso while performing the activity; or that the true center of the humeral articulation did not coincide with the estimated geometric center of the humeral head. Less hysteresis was noted with the fixed-supinated forearms, possibly suggesting that a difference in the fixed forearm positions could also influence the hysteresis observed in the angle-angle profile. As this was a preliminary study, future work on other fixed forearm positions, including fixed-pronation forearms would be of interest.

In trying to understand the effect compensatory kinematics had on load transmissions across the joint; earlier studies have suggested that a "fixed" or injured forearm results in the axis of pronation-spination being altered or affected (Graham et al., 1998; Kasten et al., 2009; Nakamura et al., 1999; Oka et al., 2006; Sauerbier et al., 2002). In normal forearms, the pronator and supinator muscles are required to rotate the radius about the ulna to affect movement, and to develop a torque. In the case of a fixed-supinated or fixed-neutral forearms, the shoulder muscles—all three groups of muscles (adductors-abductors, flexors-extensors and internal and external rotators of the shoulder) would now have to function as the pronator and supinator muscles, with the whole upper extremity replicating forearm rotation (Kapandji, 2001; Kreulen et al., 2007). However, this new set of "pronators" and "supinators" would now also have to work against the weight of the whole upper limb without the benefit of having an "ulnar"-like structure to transmit the loads, proximally. The fact that these muscles are now required to pronate and supinate the hand, suggest that the axis of pronation-supination of the hand has been altered to a more global axis of rotation. The altered (helical) axis is imaginary, requiring each shoulder muscle to have a moment arm about this imaginary axis to develop a torque required to pronate or supinate the hand. The evidence for an altered "global" pronation-supination axis is the increased articulation

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at shoulder, implying that the whole upper limb was now rotating about a new or altered axis, to generate the required moment to compensate for the loss in forearm rotation. This could only mean that the shoulder muscles have to work harder. Kasten et al. (2009) in their study of radioulnar synostosis cases, which is a fixed forearm deformity, supports this argument of increase usage of the shoulder muscles. They report how a fixed supinated hand becomes awkward when using the hand to write or play the piano and how it becomes extremely fatiguing because of the increased shoulder abduction and internal rotation required to compensate for forearm pronation (Kasten et al., 2009).

The increased range of motion to compensate for loss of forearm rotation is also suspected to change the transmission of forces across the joint. Although not calculated in this study, if we were to consider a case of a fixed-supinated forearm attempting to pronate the hand, the likelihood is that glenohumeral joint contact forces would increase with increased abduction. We can infer this from past studies by Poppen and Walker (1978) and Terrier et al. (2007), where increased gleno-humeral joint forces were recorded with increased shoulder abduction. We can also infer that joint contact forces can further increase with an externally applied load at the distal end, i.e., as in carrying a load (Poppen & Walker, 1978; Terrier et al., 2007). This would support the argument that compensatory shoulder kinematics in fixed supinated forearms could be prone to other risk of injuries.

It is important to point out that this study did have several other technical limitations. Firstly, there are only a few studies which look at the kinematics of the upper limb and its function and only recently has there been an interest in the establishing standards for motion analysis of the upper extremities. There are even fewer studies looking at the in-vivo kinematics of an upper limb with injuries or disorders (Anglin & Wyss, 2000; Rau et al., 2000; Wu et al., 2005). This made it difficult to find comparative data and led us to simulate in normal subject a restricted forearm rotation by using forearm splints. The other limitation was that the sample size used was small and resulted in some variability in the measures recorded. We are uncertain if the variability was due to the gender or anthropometric differences. In our cohort, their forearm length (measured from the distal ulna styloid to the olecranon) ranged from 28 to 30 mm, which allowed for the same splints to be used. Gender differences might influence the joint contact forces and load transmissions measured, but as these we not done in our study, we assumed that the kinematics measures were not influenced by the gender difference given similar anthropometry measures in our cohort of subjects. Our purpose of the study was to establish a method for measuring the in-vivo compensatory shoulder kinematics in a simulated loss of forearm rotation and to develop a baseline for comparison. Our intention for future work is to review cases such as posttraumatic malunited forearm fractures, distal radius fractures, radioulnar synostosis which have been reported to have restrict forearm rotation (Cooney et al., 1980; Kasten et al., 2009; Linscheid, 1992). The practicality of the study is to allow us to assess the performance of patients with forearm injuries, in doing various unilateral activities of daily living which require forearm rotation, and how we can compare and quantify their progress at different stages of recovery or repair. Comparisons of changes in joint range of motions when doing various tasks may also provide a tool to predict the risk of injuries at the shoulder and elbow (Van Andel et al., 2008; Kasten et al., 2009). In trying to establish an in-vivo kinematic model for our laboratory setting, we felt that the recently reported markers sets for motion analysis of the upper limb which considered the forearm as a single segment was not sufficient if we were to assess patients with a spectrum of forearm disorders. So in this study, we establish an additional segment to understand the rotation of the radius about the ulna. This we felt would allow us to apply the model to cases like a malunited forearm fracture or a disorder involving a discontinuity of either the radius or ulna (Nakamura et al., 1999; Tay et al., 2008). To measure forearm rotation we were also aware that the skin close to the antero-proximal joint rotated less around the longitudinal axis compared with that at the distal end of the forearm; and these might contribute to errors in the results (Cutti et al., 2006). Hence we introduced additional markers at the distal half of the radius, and at postero-proximal end of the ulna, at the olecranon. This we felt was justified, as we wanted to compare the forearm performance of each subject doing the 5 tasks under different conditions. The data were threfore presented with respect to the normal forearm as the control group (Boone & Azen, 1979; Morrey et al., 1981; Safaee-Rad et al., 1990). The other challenge was to measure the articulation about the shoulder. In this paper we deviated from the ISB recommendations applying the floating axis calculations to determine a more clinically relevant description. Abduction-adduction, reverse-forward flexion, and internal-external rotation were used to allow us to have a clear idea as to the muscle groups that were involved. Our purpose for using Grood and Suntay's equations in our laboratory was also to determine the translations of each segment at the joint (although this data were not presented in this study).

In summary, in spite of these limitations and given that the upper limb is a series of interconnected segments involving several joints, the study quantified that with forearms restricted in their range of forearm rotation the shoulder is used to compensate function when performing unilateral tasks that require mandatory prosupination of the forearm. This would increase the usage at the shoulder and would support the proposition, that an assessment of the overall movement strategy of the upper extremity is necessary, particularly in planning rehabilitation therapies (Kreulen et al., 2007).

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